Schedule

Practice problems: on LearnHub

Sample case today

Case Exam 1: Saturday, April 19th at 18:15-20:45 in labs SNAB149 and SNAB242

- Seating charts will be announced
- Only lab computers can be used
- Safe Exam Browser; Phyton, Excel, common report area

Midterm: Friday, April 25th at 19:00-21:00, on LearnHub, multiple choice

- Safe Exam Browser, only Excel can be used
- Seating charts will be announced

1

Main idea:

Collect a random sample

Determine the parameter of interest, say θ

Find a random variable, say W, that involves parameter θ and whose distribution you can specify

Use the distribution of W to find random lower and upper bounds on θ

2

Parameter of interest: true mean, μ

- $X_1, \ldots, X_n$: A random sample with $E[X] = \mu$
- We are looking for a random variable $Y=h(X_1,\ldots,X_n; \mu)$ whose distribution can be specified.

	X is normal		X is not normal		
	n<40	n≥40	n<30	30≤n<40	n≥40
σ: known	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	?	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$
σ: unknown	$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$	$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}}$	?	? $T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$?	$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}}$

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CIs around the true mean, μ

- $X_1, \ldots, X_n$: A random sample with $E[X] = \mu$
- We are looking for a random variable $Y=h(X_1,\ldots,X_n; \mu)$ whose distribution can be specified.

	X is normal		X is not normal		
	n<40	n≥40	n<30	30≤n<40	n≥40
σ: known	$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$	?	$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$
σ: unknown	$\bar{x} \pm t_{\alpha/2}\frac{s}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2}\frac{s}{\sqrt{n}}$	?	? $\bar{x} \pm t_{\alpha/2}\frac{s}{\sqrt{n}}$?	$\bar{x} \pm z_{\alpha/2}\frac{s}{\sqrt{n}}$

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CIs around the true population proportion p

X: Number of successes in n trials, each trial is i.i.d. with constant p=P{success}

We are looking for a random variable , Y=h(X; θ) whose distribution can be specified.

$\hat{P} = \frac{X}{n}$

$n\hat{p}\geq 10$ and $n(1-\hat{p})\geq 10$	**No condition on n or p**
$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	$\tilde{p} \pm z_{\alpha/2}\sqrt{\frac{\tilde{p}(1-\tilde{p})}{\tilde{n}}}$

$\tilde{P} = \frac{X+2}{n+4}$

$\tilde{n} = n+4$

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Small-Sample Confidence Intervals for the Difference Between Two Means

$X_1,\ldots,X_n$: a random sample from a population with μ_X
$Y_1,\ldots, Y_n$: a random sample from a population with μ_Y

$100(1-\alpha)\%$ CI for the difference $\mu_X - \mu_Y$ is when $n_X = n_Y = n$

$$\bar{X} - \bar{Y} \pm t_{v,\alpha/2}\sqrt{\frac{s_X^2 + s_Y^2}{n}} \qquad v = (n-1)\left(1 + \frac{2S_1^2S_2^2}{S_1^4 + S_2^4}\right)$$

Independent samples

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Small-Sample Confidence Intervals for the Difference Between Two Means - Paired

$X_1,\ldots,X_n$: a random sample from a population with μ_X

$Y_1,\ldots, Y_n$: a random sample from a population with μ_Y

$D_1= X_1 - Y_1,\ldots, D_n= X_n - Y_n$: a random sample from a population with μ_D

$100(1-\alpha)\%$ CI for the difference $\mu_D = \mu_X - \mu_Y$ is

$$\bar{D} \pm t_{n-1,\alpha/2}\sqrt{\frac{s_D^2}{n}}$$

Positively correlated samples

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Inferences about the true variance of a random variable X

$X_1, \ldots, X_n$: A random sample

We are looking for a test statistics, $Y=h(X_1,\ldots,X_n; \theta)$ such that its probability distribution does not depend on θ or any other unknown parameters.

X~Normal		X is not normal	
n<40	n≥40	n<40	n≥40
$\chi^2_{n-1} = \frac{(n-1)S^2}{\sigma^2}$	$\chi^2_{n-1} = \frac{(n-1)S^2}{\sigma^2}$	—	—

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Prediction interval (PI) for a single future value

$X_1,\ldots,X_n$: N(μ, σ) random sample with unknown μ and σ.

Goal: to predict a single value of a future X, which is a random variable

In CI, goal: to estimate the mean value of that random variable, which is unknown but constant

$$\left(\bar{x} \pm t_{n-1,\alpha/2} s \sqrt{1+\frac{1}{n}}\right)$$

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Tolerance Intervals

Let $X_1, \ldots, X_n$ be a sample from a *normal* population.

A tolerance interval for containing $100(1-\gamma)\%$ of the population with confidence $100(1-\alpha)\%$ is:

$$\bar{X} \pm k_{n,\alpha,\gamma} s$$

↓

Table A.4

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Class Exercise

A manufacturer produces a certain type of battery. The quality standards require the capacity of these batteries to be in between (137, 143) ampere-hours.

To meet these standards, the production process managers decided to design the production process such that:

- Let μ be the population average of the battery capacities. The production process should ensure that $\mu = 140$.
- Let σ^2 be the population variance of the battery capacities. The production process should ensure that $\sigma^2 = 4$.
- Let p be the proportion of batteries that meet the quality standards. The production process should ensure that $p = 0.95$.

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Class Exercise

An independent consumer-protection agency wishes to test whether the manufacturer complies to the quality standards. For this purpose, they measure the capacity of a random sample of 20 batteries from a recently produced batch. They compiled the following information from this sample.

Note that the random sample consists of $(X_1, X_2, \ldots, X_n)$:

- X_i denotes the capacity of ith battery.
- Y denotes the number of batteries complying the quality standards.
- n is the sample size.

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Does the manufacturer meet the quality standards? Why or why not? Explain.

Class Exercise

Parameter	μ	σ^2	p
Point estimator	Sample capacity mean, $\bar{X}$	Sample capacity variance, S^2	$\hat{P} = \frac{Y}{n}$
Point estimate	$\bar{x} = 138.47$	$s^2 = (2.66)^2 = 7.08$	$\hat{p} = \frac{y}{n} = 0.65$
Standard error of the point estimator	$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$	No closed-form expression	$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$
Estimated standard error of the point estimator	$\hat{\sigma}_{\bar{X}} = \frac{s}{\sqrt{n}} = 0.595$	No straightforward computation	$\hat{\sigma}_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.11$
Expression for 95% CI	$(\bar{x} \mp t_{0.025,19}\hat{\sigma}_{\bar{X}})$	$\left(\frac{(n-1)s^2}{\chi^2_{n-1,\alpha/2}}, \frac{(n-1)s^2}{\chi^2_{n-1,1-\alpha/2}}\right)$	$(\hat{p} \mp z_{0.025}\hat{\sigma}_{\hat{p}})$
95% CI	(137.23, 139.71)	(4.09, 15.08)	(0.44, 0.86)
95% PI	(135.16, 141.78)	--	--
95% TI for 90%	(123.32, 153.62)	--	--

The quality standards require the capacity of these batteries to be in between (137, 143) ampere-hours.

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Does the production process satisfy the managers' design? Why or why not? Explain.

Class Exercise

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95% PI	(135.16, 141.78)	--	--
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Managers' design →

The production process should ensure that μ = 140.
The production process should ensure that σ^2 = 4.
The production process should ensure that p = 0.95.

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Class Exercise

If the production process satisfied the managers' design, do you think it would also meet the quality standards? Why or why not? Explain.

Parameter	μ	σ^2	p
Point estimator	Sample capacity mean, $\bar{X}$	Sample capacity variance, S^2	$\hat{P} = \frac{Y}{n}$
Point estimate	$\bar{x} = 138.47$	$s^2 = (2.66)^2 = 7.08$	$\hat{p} = \frac{y}{n} = 0.65$
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The quality standards require the capacity of these batteries to be in between (137, 143) ampere-hours.

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The production process should ensure that $\sigma^2 = 4$.
The production process should ensure that $p = 0.95$.

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Any questions on confidence intervals?

Now Hypothesis Testing…

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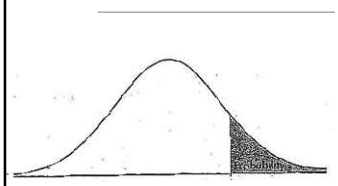


TABLE B: *t*-DISTRIBUTION CRITICAL VALUES

	Tail probability p											
df	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	.765	.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	.741	.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	.727	.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	.718	.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	.711	.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	.706	.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	.703	.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	.700	.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	.697	.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	.695	.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	.694	.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	.692	.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	.691	.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	.690	.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	.689	.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	.688	.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.611	3.922
19	.688	.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	.687	.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	.686	.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	.686	.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	.685	.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	.685	.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	.684	.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	.684	.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	.684	.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	.683	.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	.683	.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	.683	.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646
40	.681	.851	1.050	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551
50	.679	.849	1.047	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496
60	.679	.848	1.045	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	.678	.846	1.043	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	.677	.845	1.042	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390
1000	.675	.842	1.037	1.282	1.646	1.962	2.056	2.330	2.581	2.813	3.098	3.300
∞	.674	.841	1.036	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.091	3.291

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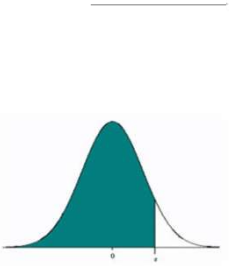


Table of Standard Normal Probabilities for Positive Z-scores

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

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χ^2-table

Chi-Square (χ^2) Distribution

Degrees of Freedom	Area to the Right of Critical Value							
	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01
1	—	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277
5	0.554	0.831	1.145	1.610	9.236	11.071	12.833	15.086
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209
11	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725
12	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217
13	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688
14	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141
15	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578
16	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000
17	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409
18	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805
19	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191
20	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566
21	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932
22	9.542	10.982	12.338	14.042	30.813	33.924	36.781	40.289
23	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638
24	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980
25	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314
26	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642
27	12.879	14.573	16.151	18.114	36.741	40.113	43.194	46.963
28	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278
29	14.257	16.047	17.708	19.768	39.087	42.557	45.722	49.588
30	14.954	16.791	18.493	20.599	40.256	43.773	46.979	50.892